NOTICE OF CHANGE

METRIC

DOD-STD-1399(NAVY)
SECTION 533
NOTICE 1
13 April 1990

MILITARY STANDARD
INTERFACE STANDARD FOR
SHIPBOARD SYSTEMS
SECTION 533
POTABLE WATER SERVICE
(METRIC)

TO ALL HOLDERS OF DOD-STD-1399(NAVY), SECTION 533:

1. THE FOLLOWING PAGES OF DOD-STD-1399(NAVY), SECTION 533 HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
ii	13 April 1990	ii	22 May 1978
1	13 April 1990	1	22 May 1978
1A	13 April 1990	1	22 May 1978
2	13 April 1990	2	22 May 1978
2A	13 April 1990	2	22 May 1978
3	13 April 1978	3	22 May 1978
4	13 April 1990	4	22 May 1978
5	13 April 1990	5	22 May 1978
6	13 April 1990	6	22 May 1978

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of DOD-STD-1399(NAVY), Section 533 will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

Preparing activity:
Navy – SH
(Project 1990-N081)

AMSC N/A

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DEPARTMENT OF THE NAVY

WASHINGTON, DC 20362-5101

Interface Standard for Shipboard Systems,
Potable Water Service

DOD-STD-1399(NAVY)
SECTION 533

1. This Military Standard is approved for use by all interested Commands of the Department of the Navy in the technical development plans, design, and procurement specifications for new ship acquisitions, ship modernizations or conversions, and equipment for installation therein and into active fleet ships where applicable, and is available for use by all departments and agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z33, Department of the Navy, Washington, DC 20362-5101, by using the self-address Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. GENERAL, SCOPE, INTERFACE, AND APPLICABILITY

1.1 General. The policies and procedures established by DOD-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document.

1.2 Scope. This section establishes interface requirements for shipboard equipment utilizing potable water services to ensure compatibility between such equipment and the potable water service system.

1.3 Interface. The basic characteristic and constraint categories concerned with this interface are shown symbolically on figure 1 (see section 3, "Definitions" of DOD-STD-1399):

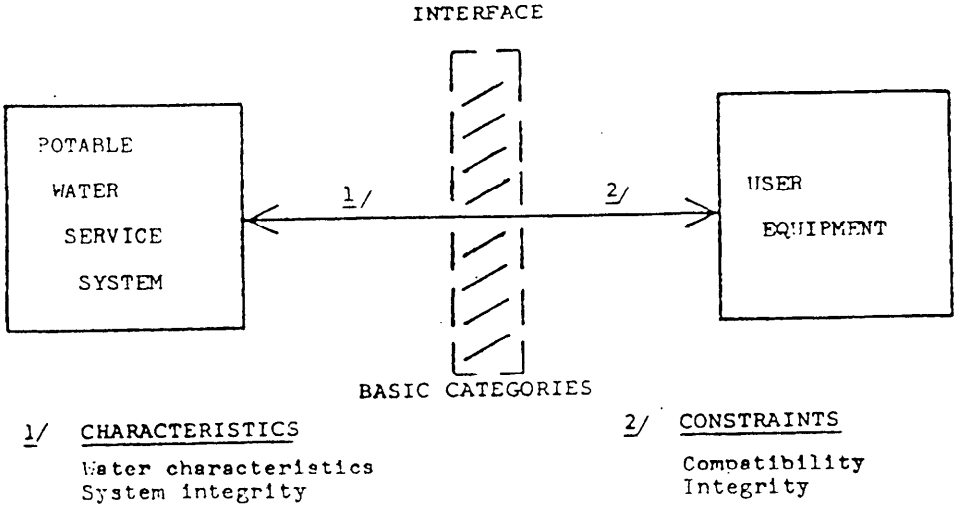


FIGURE 1. *Interface.*

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

1.4 Applicability. This section is directly concerned with the design and installation considerations at the interface between the potable water system and its users which affect the overall compatibility at the interface. For criteria concerning related fresh water interface areas, see other pertinent sections of DOD-STD-1399 in the 530 series.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on the date of invitation for bids or request for proposal forms a part of this standard to the extent specified herein.

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PUBLICATION

MILITARY

NAVY MEDICAL COMMAND (NAVMED)
NAVMED P-5010 – Manual of Naval Preventative Medicine.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Supersedes page 1 of 22 May 1978
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3. DEFINITIONS

3.1 Potable water. Potable water is water whose bacteriological quality and physical and chemical characteristics make it suitable for human consumption.

3.2 Potable water service system. A potable water service system is a shipboard system, including the necessary tank filling, storage, transfer, disinfection, and service arrangement for the distribution of potable water throughout the ship.

3.3 User **equipment.** A user equipment is any facility or apparatus which utilizes or consumes water from the potable water service system.

4. REQUIREMENTS

4.1 Requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to by SYSCOMs, project managers, contractors, and all other engaged in any aspect of shipboard potable water service system design to which these requirements and constraints apply, including equipment design, production, and installation (see section 4, "Requirements" of DOD-STD-1399).

5. INTERFACE CHARACTERISTICS AND CONSTRAINTS

5.1 General considerations. The potable water service system provides for a continuous supply of potable water from the storage tanks to the various users throughout the ship. The system provides cold water directly to users, and provides water to heaters for hot water needs. The storage tanks receive water from shore, from another ship or from the ship's distilling plants. Except on nuclear submarines, the water will be disinfected by bromine disinfection systems. Figure 2 shows the basic elements of a typical potable water service system.

5.2 Interface characteristics. The interface characteristics of the ship's potable water service system are described in 5.2.1 through 5.2.4.

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5.2.1 **Water quality standards.** When water enters the potable water service system, it is assumed to be of a satisfactory quality with respect to the bacteriological content and physical and chemical characteristics. Water quality standards and the criteria for their application are given in NAVMED P-5010.

5.2.2 **Pressure.** The water pressure supplied will be within the range of 35 kilopascals (kPa) (5 pounds per square inch (lb/in^2)) to 700 kPa (100 lb/in^2).

5.2.3 **Temperature.** The temperature of the water supplied will be as follows:

a. Cold potable water – 2 °C to 32 °C (35 °F to 90 °F)
b. Hot potable water – 52 °C to 57 °C (125 °F to 135 °F).

5.2.4 **Bromine chlorine residual.** A trace of free bromine or chlorine residual (less than 0.2 milligrams per liter (mg/L)) will be maintained in the potable water distribution system at all times, except on nuclear submarines (see 20.5 of the Appendix).

5.3 **Interface constraints.** The interface characteristics of the ship's potable water system impose certain constraints on the design and installation of equipment utilizing or consuming potable water. These constraints are specified in 5.3.1 through 5.3.3.

5.3.1 **Compatibility.** The design and installation of equipment utilizing potable water service shall be compatible with the interface characteristics given in 5.2.

5.3.2 **Integrity of the potable water system at a delivery point.** When hot or cold potable water is delivered to user equipment, the potable water system shall be protected from contamination.

5.3.2.1 **Air gaps.** Where portable water is delivered to other systems, tanks, or facilities that could contain water of inferior quality, an air gap of at least two supply pipe diameters shall be provided between the supply and receiving connections. A funnel shall be provided in the air gap receiving connection wherever necessary. Location of faucets above the rims of lavatories, sinks, and other open receiving vessels constitutes an adequate air gap. Some examples of facilities which require and air gap are: galley and pantry sinks, ice machines, dishwashers and other scullery equipment, vegetable peelers, steam kettles, laundry machines, and sterilizers.

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5.3.2.2 **Hose connections.** Where potable water is delivered to a user equipment by means of a hose connection, the connection shall be fitted with a valve of the stop-check type and a vacuum breaker back flow preventer.

5.3.2.3 **Permanent connections.** Where potable water is permanently connected to a system or component that could pressurize the potable water system with contaminants, a reduced pressure backflow preventer will be installed.

5.3.3 **External filling and delivery connections.** The connections for receiving potable water from external sources, or delivering water to another ship shall be:

a. *Surface ships:* 63 millimeters (mm) (2-1/2 inch) hose value connection (except for ships with a rated capacity of less than 50 gallons per minute, where the hose value connection shall be 38 mm (1-1/2 inch)) located on the weather deck.

b. *Submarines:* 38 mm (1-1/2 inch) hose valve connection located within the pressure hull.

Such hose valve connections shall be painted dark blue, provided with a cap and chain, and plainly marked "POTABLE WATER ONLY".

6. DEVIATIONS

6.1 Conditions. In achieving the purpose of this section, it is recognized that there must be some flexibility of application. During the early design stage of shipboard equipment utilizing potable water service, it may become apparent that significant advantages in the overall design/operation of such equipment can be achieved by deviating from the standard characteristics specified herein. In such instance, the provisions of section 6, "Deviations" of DOD-STD-1399 shall be complied with.

6.1.1 Deviation procedure. Requests for deviations shall be submitted to the Naval Sea Systems Command with copies to:

a. Program/project manager
b. NAVSEA 56Y3

Review activity:
Navy - YD

User activities:
Navy - EC, AS

Preparing Activity:
Navy - SH
(Project 1990-N022)

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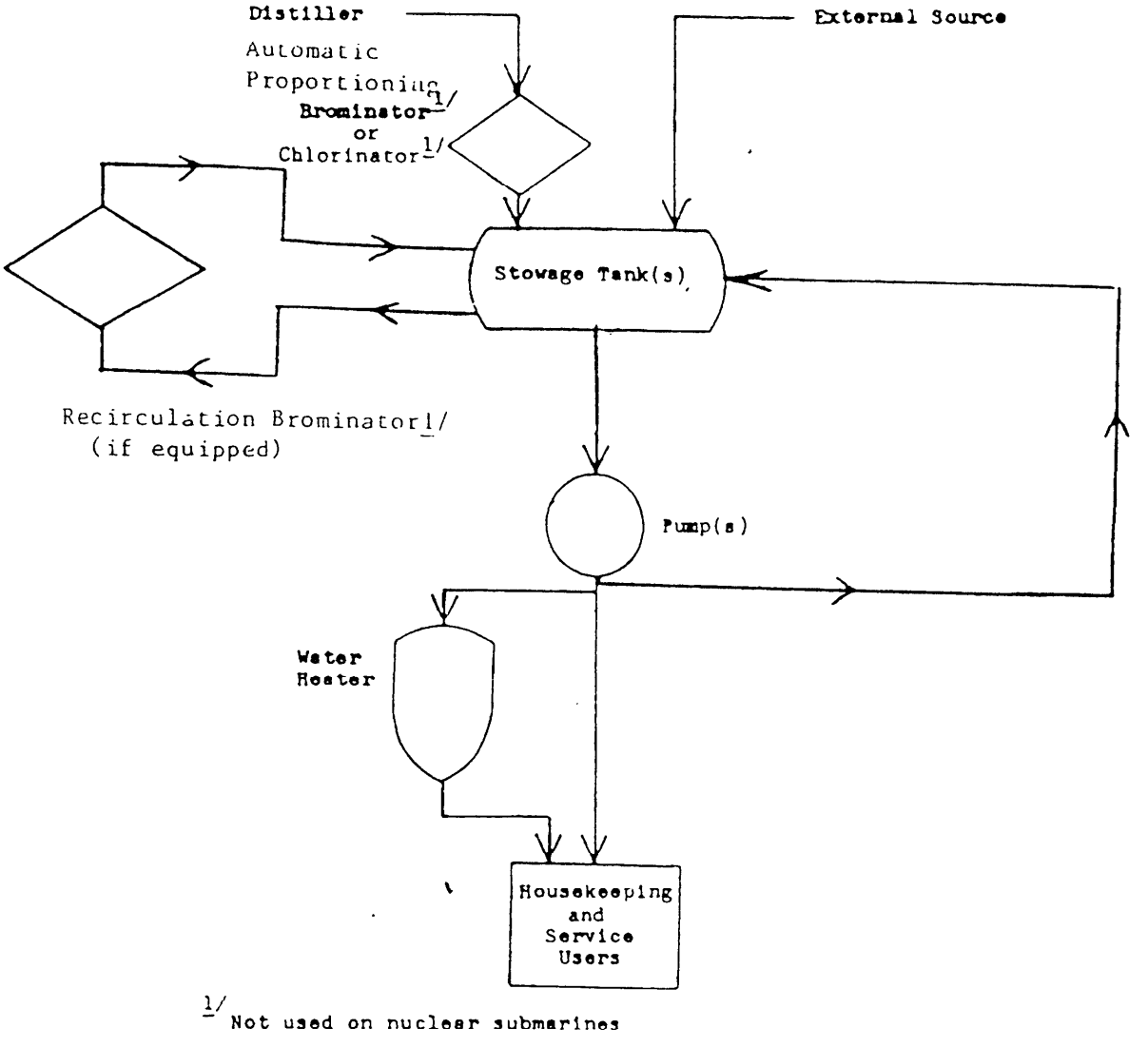


FIGURE 2. *Basic elements, typical potable water service system.*

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